

AYUSH LODH

Curriculum Vitae

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Work Experience

- Feb, 2024 – **Research Intern**, *Computer Vision and Pattern Recognition Unit*, Indian Statistical Institute, Kolkata, India.
April, 2024 Worked on online and offline handwritten text recognition from a webcam or camera live feed.
- May, 2024 – **Junior AI Engineer**, *Habitat Lens Private Limited*, Kolkata, India, AI Solutions.
July, 2024 Gained comprehensive exposure to Handwritten Document Images, Annotation Task
- Dec, 2025 – **Research Intern**, *Computer Vision Center*, Universitat Autònoma de Barcelona, Spain (Remote).
Mar, 2026 Worked on Multimodal RAG using Graph Neural Networks on Enterprise Document Images

Education

- 2025 – **Master of Technology, Computer Science Engineering (Analytics)**, *National Institute of Technology*, Delhi, India. CGPA: 7.5/10.
Present **Courses:** Data Mining and Warehousing, Deep Learning, Advanced Databases
Ongoing Project: From Chunks to Graphs, *Advisors: Prof. Dr. Nisha Singh Chauhan, NIT Delhi & Dr. Sanket Biswas, CVC, UAB*
- 2020 – 2024 **Bachelor of Engineering, Information Technology**, *University Institute of Technology*, Burdwan University, India. CGPA: 8.79/10.
Courses: Image Processing, Artificial Intelligence, Machine Learning
Final Year Project: 3D - Playfair Cipher with Dual Key, *Advisor: Ms. Nabanita Das*
- 2018 – 2020 **Higher Secondary Examination**, *D.A.V. Public School, Central Board Of Secondary Education*, Asansol, Paschim Bardhaman, WB, India, Percentage: 81.8%.
- 2018 **Secondary Examination**, *D.A.V. Public School, Central Board Of Secondary Education*, Asansol, Paschim Bardhaman, WB, India, Percentage: 84.2%.

Publications

In Conference Proceedings

- 2026 **Ayush Lodh**, Souparni Mazumder, Sanket Biswas, Josep Lladós, and Nisha Singh Chauhan. From chunks to graphs: Training-free multimodal late interaction for document understanding. In *International Conference on Document Analysis and Recognition*, pages 592–609. Springer, 2026.
- 2026 Alloy Das, Sanket Biswas, Soumitri Chattopadhyay, **Ayush Lodh**, Aniket Pal, Priyanka Banerjee, Nisha Singh, CV Jawahar, Josep Lladós, and Dimosthenis Karatzas. Notes-bank: Benchmarking vision-language models on unstructured academic and scientific note-taking. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 700–718. Springer, 2026.
- 2025 **Ayush Lodh**, Ritabrata Chakraborty, Shivakumara Palaiahnakote, and Umapada Pal. A transformer based handwriting recognition system jointly using online and offline features. In *Asian Conference on Pattern Recognition*, pages 250–264. Springer, 2025.

- 2025 Aniket Pal, Sanket Biswas, Alloy Das, **Ayush Lodh**, Priyanka Banerjee, Soumitri Chattopadhyay, Ajoy Mondal, Dimosthenis Karatzas, Josep Lladós, and CV Jawahar. ICDAR 2025 handwritten notes understanding challenge. In *International Conference on Document Analysis and Recognition*, pages 553–567. Springer, 2025.

Research Experience

Indian Statistical Institute, Kolkata, India.

- Feb, 2024 – **A Transformer Based Handwriting Recognition System Jointly Using Online and Offline Features**, Advisor: Prof. Dr. Umapada Pal.

Proposed an end-to-end transformer for writer-independent handwriting recognition that fuses offline image patches and online stroke tokens via learnable latent queries in a shared space.

Habitat Lens Private Limited, Kolkata, India.

- May 2024 – **NoTeS-Bank: Benchmarking Neural Transcription and Search for Scientific Notes Understanding**, Advisor: Mr. Alloy Das.

Introduced a benchmark for transcription and search over scientific handwritten notes with mixed equations, diagrams, and text, defining Evidence-Based and Open-Domain VQA tasks for vision–language models.

Universitat Autònoma de Barcelona, Spain

- Dec, 2025 – **From Chunks to Graphs**, Advisors: Dr. Sanket Biswas & Dr. Josep Lladós.

Mar, 2026 Developed a training-free multimodal retrieval framework replacing chunk-based selection with structure-aware subgraph retrieval over Doc2Graph representations, improving factual consistency on document VQA over text-RAG baselines.

Skills

- Programming Python, C/C++
Libraries PyTorch, Tensorflow
Documentation MS Office, $\text{\LaTeX}$
Languages English, Bengali (Native) & Hindi